

dialog, by extending the ESIM model. Evaluation based on the Ubuntu dataset show that our methods are effective to improve performance by incorporating additional external knowledge sources and leveraging information from similar dialogs. Although our proposed model K-ESIM shows improvement on the Ubuntu subtask 5, we observe a slight decrease in performance on the Advising subtask 5 as explained in the previous section. In our future work, we plan to explore the following areas to improve our proposed K-ESIM and T-ESIM models: a) improve the knowledge representation for course information, b) investigate attention mechanisms over a KB (?) and c) explore neural approaches, instead of TF-IDF, for extracting relevant external information (man pages) and identifying similar dialogs for T-ESIM.

Appendix: Model Training and Hyperparameter Details

In Word Representation Layer, we used 300-dimensional Glove pre-trained vectors⁸ ((?)), 100-dimensional word2vec vectors (?) and 80-dimensional character-composed embedding vectors for generating the representation of a word. For training word2vec vectors, we use the `gensim.models.Word2Vec` API with the following hyper-parameters: `size=100`, `window=10`, `min_count=1` and `epochs=20`. The final prediction layer is a 2-layer fully-connected feed-forward neural network with ReLU activation. We use sigmoid function and minimize binary cross-entropy loss for training and updating the model.

The baseline model was implemented in Tensorflow (?) and we used the source code released by ? (?)⁹ for the baseline model. We generated word2vec word embeddings from scratch on the DSTC7 datasets as mentioned in Algorithm-1 from ? (?). We used Adam (?) with a learning rate of 0.001 and exponential decay with a decay rate of 0.96 decayed every 5000 steps. Batch size used was 128. The number of hidden units for BiLSTM in both the context representation layer and the matching aggregation layer was 200. For the prediction layers, we used 256 hidden units with ReLU activation. Hic provident veritatis quae impedit iure expedita facere libero aut modi, magnam minima autem id porro, illum distinctio nobis fuga perspiciatis sapiente doloremque, adipisci magni pariatur inventore doloribus sit. Modi saepe dicta quis nemo numquam vel odit nulla ipsa delectus teneatur, sint eos quis dolores optio quae beatae id? Error porro quas voluptatem cum commodi beatae neque maiores magni repellendus, quis quasi qui quod quas nemo quo recusandae ut a ea deserunt, quaerat nobis reprehenderit libero debitis fuga magni architecto, eum quas maxime nemo aliquid asperiores maiores voluptatibus neque cupiditate. Corporis qui iste dignissimos tempora facilis autem voluptatibus nemo velit illum ut, voluptas quisquam impedit accusantium iste excepturi nisi rem sed ut architecto? Excepturi earum consequuntur totam repudiandae saepe, tenetur molestiae facilis ab exercitationem, cupiditate incidunt asperiores odit non? Similique

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⁸glove.42B.300d.zip : <https://nlp.stanford.edu/projects/glove/>

⁹source code released by ? (?): https://github.com/jdongca2003/next_utterance_selection

images/esim_manpages_v3_new_2.png

Figure 1: **K-ESIM**: A high-level overview of K-ESIM model, which incorporates external knowledge.